

S M Hasan Mansur

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Nacogdoches, Texas, USA

SUMMARY

Assistant Professor in CS at Stephen F. Austin State University. **Research Interests:** Machine Learning and Deep Learning with applications in Software Engineering and HCI. **Publication** track record in top-tier SWE venues. 5+ years of **professional software engineering** experience.

EDUCATION

- **George Mason University** August 2026
PhD in Computer Science Virginia, USA
 - Dissertation: Automated User Interface Design and Understanding through Multimodal Artificial Intelligence
 - Advisor: [Dr. Kevin Moran](#)
- **George Mason University** May 2023
MS in Computer Science Virginia, USA
 - CGPA: 3.83/4.00
- **Shah Jalal University of Science & Technology** April 2007
BS in Computer Science Sylhet, Bangladesh
 - CGPA: 3.54/4.00

TEACHING & RESEARCH EXPERIENCE

- **Stephen F. Austin State University** September 2026 - Present
Assistant Professor, Computer Science Texas, USA
 - Teach undergraduate courses in Computer Programming Principles and Object-Oriented Programming Methods, emphasizing Java programming, computational thinking, and software development.
 - Design and deliver lectures, coding demonstrations, programming assignments, projects, quizzes, and examinations aligned with course learning outcomes.
 - Develop hands-on programming activities and projects to strengthen students' algorithmic thinking, problem-solving, debugging, and software development skills.
 - Assess student learning through programming assignments, projects, quizzes, examinations, and individualized feedback.
 - Mentor and support students through office hours and one-on-one guidance on programming concepts, debugging, and software development practices.
- **George Mason University** May 2025 - May 2026 | May 2021 - August 2024
Graduate Research Assistant Virginia, USA
 - Major areas of contribution: Software Documentation, Ethical Software Design, Software Accessibility.
 - Lead the project *MultiC2C* that aims to benchmark the performance of state-of-the-art multimodal LLMs in generating context-aware summaries from code snippet, related UI screenshot and metadata.
 - Designed and developed *AidUI*, a novel approach to detect and localize deceptive design patterns on UIs.
 - Collaborated on curating dataset and developing semantic text matching prototype in project *MotorEase*, an approach to detect motor-impairment accessibility violations in app UIs.
 - Collaborated in developing benchmark datasets for *Software Evolution*, *Bug Reporting* and *Bug Localization*.
- **George Mason University** August 2024 - May 2025 | August 2018 - May 2021
Graduate Teaching Assistant Virginia, USA
 - Courses: Python Programming, Database, Computer Vision, Formal Methods.
 - Helped design programming assignments and evaluation rubrics.
 - Prepared and conducted lab sessions, tutorials, and recitations.
 - Graded assignments, quizzes, and exams with consistent and timely feedback.
 - Managed course forums and provided one-on-one academic assistance to students through regular office hours.

PROFESSIONAL EXPERIENCE

- **Ice9 Ltd.**
Software Engineer

 - Core engineering team member for developing and maintaining *SMASHBOARD.CO*, a CRM & analytics platform.
 - Collaborated on strategic planning for architecture design, development, testing and deployment.
 - Optimized database and implemented master-replica architecture to resolve scalability and synchronization issues.
 - Lead the migration of the core product from Rackspace to AWS.

April 2017 - July 2018
Dhaka, Bangladesh
- **Synchronous ICT**
Software Engineer

 - Developed user management, search and notification platforms of *COMX*, an in-house application framework.
 - Assessed team's capacity to deliver, formed sprints and released duration based development milestones.
 - Established software engineering practices like version controlling, issue tracking and test driven development.

February 2015 - April 2017
Dhaka, Bangladesh
- **Dcastalia**
Software Engineer

 - Developed *HE BUSINESS MANAGER*, an Android application for inventory management of healthcare products.

August 2013 - January 2015
Dhaka, Bangladesh
- **Mercedes-Benz Group AG (former Daimler AG)**
Student Intern

 - Developed *Key-Value* pair parser to convert the unstructured environment perception data into a canonical form.

October 2012 - July 2013
Ulm, Germany
- **Grameenphone Ltd.**
System Engineer

 - Member of the transmission network configuration and optimization team to serve over 20 million mobile users.

July 2007 - February 2011
Dhaka, Bangladesh

PUBLICATIONS

- [1] [MOBILESoft'25] Zihe Song, **S M Hasan Mansur**, Ravishka Rathnasuriya, Yumna Fatima, Wei Yang, Kevin Moran, Wing Lam, "*Can You Mimic Me? Exploring the Use of Android Record & Replay Tools in Debugging*", in Proceedings of the 2025 IEEE/ACM 12th International Conference on Mobile Software Engineering and Systems, Ottawa, Canada, April 2025.
- [2] [ICSE'24] Arun Krishnavajjala, **S M Hasan Mansur**, Justin Jose, Kevin Moran, "*MotorEase: Automated Detection of Motor Impairment Accessibility Issues in Mobile App UIs*", in Proceedings of the IEEE/ACM 46th International Conference on Software Engineering, Lisbon, Portugal, April 2024.
- [3] [ICSE'24] Junayed Mahmud, Nadeeshan De Silva, Safwat Ali Khan, Seyed Hooman Mostafavi, **S M Hasan Mansur**, Oscar Chaparro, Andrian (Andi) Marcus, Kevin Moran, "*On Using GUI Interaction Data to Improve Text Retrieval-based Bug Localization*", in Proceedings of the IEEE/ACM 46th International Conference on Software Engineering, Lisbon, Portugal, April 2024.
- [4] [MSR'24] Sabiha Salma, **S M Hasan Mansur**, Kevin Moran, Yule Zhang, "*GUIEvo: Automated Evolution of Mobile Application GUIs from Mockups*", in Proceedings of the 21st International Conference on Mining Software Repositories, Lisbon, Portugal, April 2024.
- [5] [ICSE'23] **S M Hasan Mansur**, Kevin Moran, "*Toward Automated Tools to Support Ethical GUI Design*", in Proceedings of the 2023 IEEE/ACM 45th International Conference on Software Engineering, Doctoral Symposium Track, Melbourne, Australia, May 2023.
- [6] [ICSE'23] **S M Hasan Mansur**, Sabiha Salma, Damilola Awofisayo, Kevin Moran, "*AidUI: Toward Automated Recognition of Dark Patterns in User Interfaces*", in Proceedings of the 2023 IEEE/ACM 45th International Conference on Software Engineering, Research Track, Melbourne, Australia, May 2023.
- [7] [MSR'21] Tyler Wendland, Jingyang Sun, Junayed Mahmud, **S M Hasan Mansur**, Steven Huang, Kevin Moran, Julia Rubin, Mattia Fazzini, "*AndroR2: A Dataset of Manually-Reproduced Bug Reports for Android apps*", in Proceedings of the 2021 IEEE/ACM 18th International Conference on Mining Software Repositories, Madrid, Spain, May 2021.

SELECTED RESEARCH

- **Multimodal Code Summarization**
Principal Contributor

 - Goal: To investigate on Multimodal-LLMs capabilities in capturing UI semantics for code summarization.
 - Automated a pipeline to collect and map multi-modal data (code, comments, UI) from open-source repositories.
 - Lead the team to develop ground truth code summaries by augmenting UI information.
 - Formulated the design of experiments that include different prompt techniques, input modalities and LLMs.
 - Developed the pipeline for both lexical and semantic similarity based quantitative evaluation.
 - Conducted user study to assess human evaluation of generated code summaries.

Submitted, Under Review

- **AidUI** ICSE'23
Principal Contributor [\[Paper\]](#)[\[Code\]](#)
 - *AidUI* is an automated approach to detect and localize deceptive design patterns on UIs.
 - Developed a unified taxonomy and heuristic rules to detect visual-textual cues of deceptive UI patterns.
 - Automated a data pipeline to extract UIs from public app usage videos and screenshots from prior studies.
 - Designed and developed a computer vision and NLP-based novel approach to detect deceptive UI patterns.
 - Implemented an automated evaluation pipeline. Dockerized and published research artifacts in a public repository.
- **MotorEase** ICSE'24
Collaboration [\[Paper\]](#)[\[Code\]](#)
 - *MotorEase* is an approach to detect motor-impairment accessibility violations in app UIs.
 - Collaborated with the lead contributor on labeling and curation of the dataset.
 - Developed the initial prototype of the “Semantic Text Matching” component of the automated approach.
 - Conducted baseline experiments to assess performance in touch target and icon distance detection.

TECHNICAL SKILLS

- **Programming Languages:** Python, Javascript, Java, C, PHP, HTML, Shell Scripting
- **ML/DL/NLP:** LLM (prompting, finetuning), PyTorch, Hugging Face, LangChain, scikit-learn, spaCy
- **Databases:** MySQL, MongoDB, Redis, Elasticsearch, SQLite
- **Libraries:** OpenCV, NumPy, Matplotlib, Pandas, Socket.IO
- **Backend/Cloud/DevOps:** Node.js, Express.js, REST API, AWS, Docker, Git

TEACHING TRAININGS

- **George Mason University** 2025
Graduate Teaching Academy – SkillCraft Virginia, USA
 - Title: SkillCraft Level 1 - Essentials of Student Engagement
 - Sessions: Fundamentals of Student Engagement | In Class Activities | Grading and Feedback | Classroom Climate
- **George Mason University** 2024
Stearns Center for Teaching and Learning Virginia, USA
 - Title: Inclusive Teaching for STEM Graduate Students Mini-Course
 - Sessions: Developing inclusive classroom spaces | Preparing the course | Handling classroom situations

STUDENT MENTORSHIP

- **George Mason University** 2021 - 2022
Aspiring Scientists Summer Internship Program Virginia, USA
 - High School Mentees: Damilola Awofisayo, Justin Jose

RESEARCH TALKS

- **George Mason University** April 2024
Dissertation Proposal Presentation Virginia, USA
 - Automating Software Engineering via Learning from Graphical User Interfaces
- **JPMorgan Chase & Co.** January 2024
Interview - 2024 AI Research Summer Associate Program Virtual
 - Multi-modal Learning for Software Engineering Automation
- **Los Alamos National Laboratory** January 2024
Interview - AML Summer Internship Virtual
 - Multi-modal Learning for Software Engineering Automation
- **International Conference on Software Engineering** May 2023
Research Track Melbourne, Australia
 - AidUI: Toward Automated Recognition of Dark Patterns in User Interfaces
- **International Conference on Software Engineering** May 2023
Doctoral Symposium Melbourne, Australia
 - Toward Automated Tools to Support Ethical GUI Design

HONORS & AWARDS

- **NSF Travel Award**, International Conference on Software Engineering 2023
- **Best Poster Presentation Award**, GMU CS Research Symposium 2022
- **Summer Research Initiation Award**, Dept. of Computer Science, GMU 2019

PROFESSIONAL SERVICES

- **Junior PC Member**, MSR'24 2024
- **Co-Reviewer**, MSR'22 2022
- **External Reviewer**, ICPC'21, ICSE'21, MSR'21, ASE'22, SANER'22, SANER'23, ICSE'23 2021 - 2023

LEADERSHIP & VOLUNTEER EXPERIENCE

- **Director-Web Services**, Bangladeshi Graduate Student Association, GMU 2021 - 2022
- **Member**, KIN (*Humanitarian Organization*), SUST, Bangladesh 2002 - 2007